Implicitly and Differentiably Representing Protein Surfaces and Interfaces

Author Email

Cory Scott[1] Charlie Rothschild[1] Benjamin P. Nye [1]

[1]Department of Mathematics and Computer Science, Colorado College. Equal Contribution.

Motivation

We introduce a pipeline for implicitly representing a protein, or protein complex, as the union of signed distance functions (SDFs) for each atom as van der Waals sphere. Mirroring recent successful work applying SDFs to represent 3D geometry, we present a proof of concept that this representation of proteins could be useful in several biological applications, and propose further experiments that are necessary to validate the proposed approach.

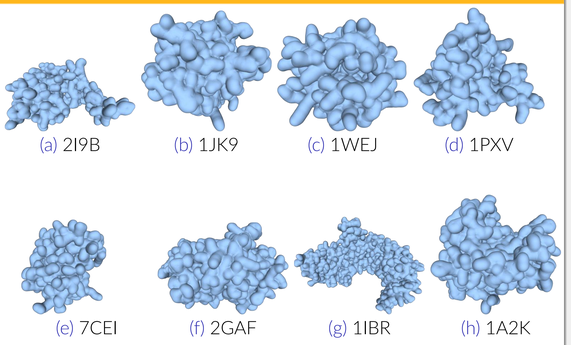
SDFs

Let $\Omega \subset \mathbb{R}^n$ be a compact subset of Euclidean space, and let $\partial\Omega$ represent its boundary. For any point $\vec{x} \in \mathbb{R}^n$, the signed distance $d_\Omega(\vec{x})$ is defined as:

$$d_\Omega(\vec{x}) = \begin{cases} 0 & x \in \partial\Omega \\ -||\vec{x} - \vec{y}|| & \vec{x} \in \Omega \\ ||\vec{x} - \vec{y}|| & \vec{x} \notin \Omega \end{cases}$$

where $\vec{y} = \arg\min_{\vec{y} \in \partial\Omega} ||\vec{x} - \vec{y}||$

Protein SDFs

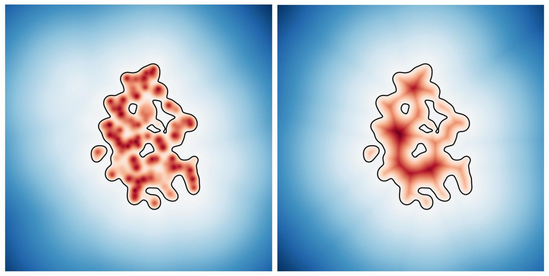


(a) 2I9B (b) 1JK9 (c) 1WEJ (d) 1PXV

(e) 7CEI (f) 2GAF (g) 1IBR (h) 1A2K

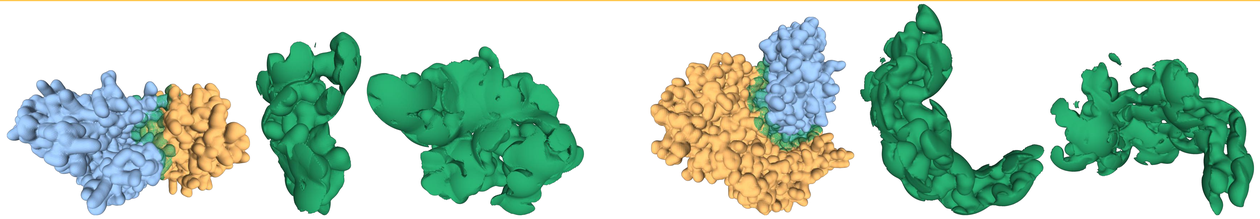
$d = 0$-level set meshes of multiple proteins from the DB5 dataset. Each protein imaged here is the "left" chain of the protein complex, in its bound confirmation.

Possible Issues



An illustration of one possible issue when representing a protein as a collection of spherical SDFs. The left shows the union SDF, while the right shows the true distance to the $d = 0$ isosurface. The union SDF is incorrect in the interior.

Protein Interface SDFs



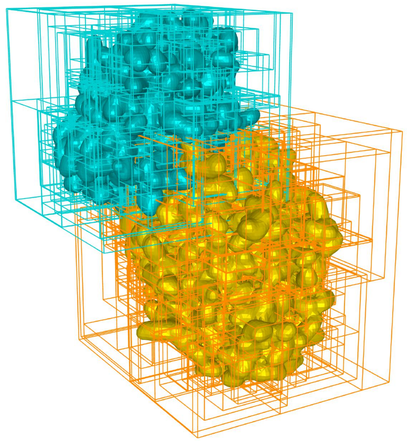
$$d_{\text{INTER}} = \min\left(\max\left(d_1(\vec{x}) - r, d_2(\vec{x})\right), \max\left(d_1(\vec{x}), d_2(\vec{x}) - r\right)\right)$$

An example of the $d(\mathbf{x}) = 0$ isosurface generated by our method for the protein 1WDW from the DB5 dataset. Left: The intersection isosurface rendered alongside both chains of the protein complex (color denotes chain ID). Nodes in the intersection region have been highlighted in green. Right: two views of the isolated intersection mesh.

Acceleration Structures

Summary

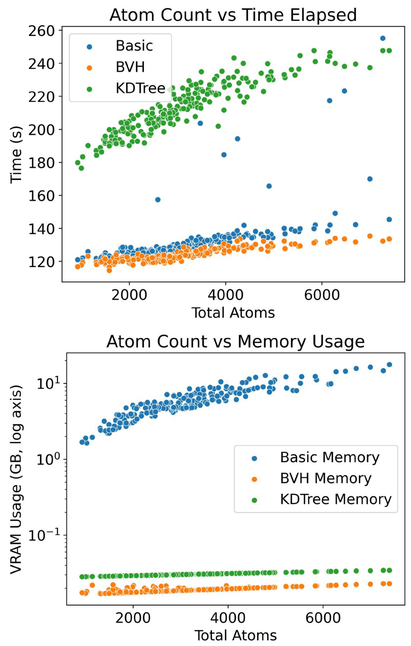
We examine two approaches for accelerating queries against proteins represented as SDFs: a Bounding Volume Hierarchy (BVH) and a KDTree of atom positions. These data structures dramatically decrease memory usage (below), at the cost of some artifacts induced at the bounaries of spatial subdivisions (right).



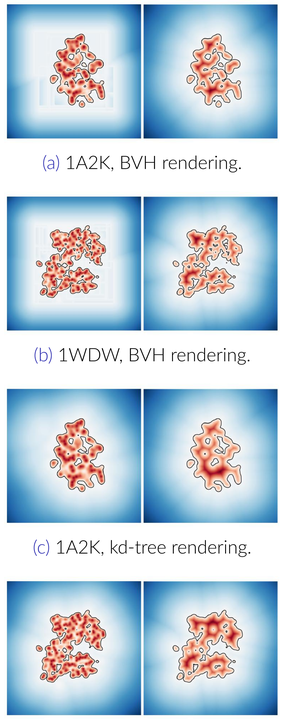
A visualization of the bounding volume hierarchy built by our method for the protein 1WDW.

	Avg time (ms)	Peak memory
Basic	48.01	1.91MB
KDTree	77.43	11.15 kB
BVH	45.77	7.06 kB

Comparison of time/memory usage by two acceleration structures.



Time/memory benchmarking of acceleration structures for querying protein SDFs.



(a) 1A2K, BVH rendering.

(b) 1WDW, BVH rendering.

(c) 1A2K, kd-tree rendering.

(d) 1WDW, kd-tree rendering.

Figure 2. Artifacts in SDFs produced by two acceleration structures.

Software and Data

Protein SDF Construction: `https://github.com/cory-b-scott/protein_sdfs`.
SDFs in PyTorch: `https://github.com/cory-b-scott/torch_sdf`.

Acknowledgments

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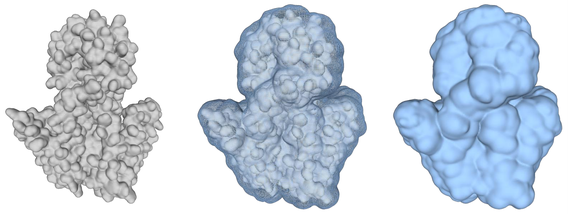
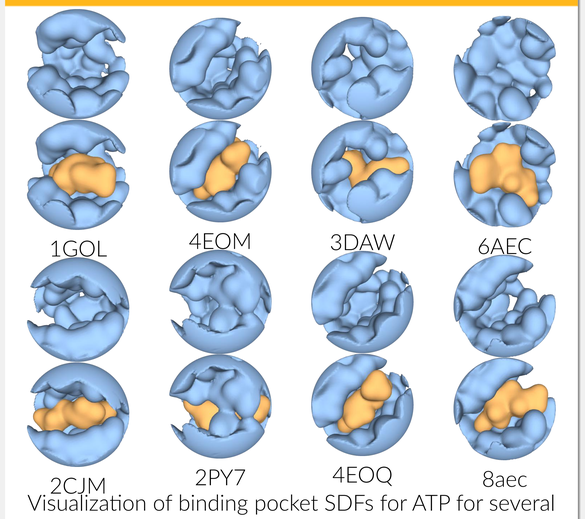
Protein Trajectory Analysis



An SDF computed over all timesteps of a MD simulation of the YiiP membrane protein in water. Left: the protein SDF for the first timestep only. Right: the SDF computed by taking the smooth-min blend of all timesteps. Center: left and right views overlaid.

Binding Pocket Visualization



Visualization of binding pocket SDFs for ATP for several proteins. Top row shows the pocket only; bottom row shows the pocket occupied by the ATP molecule (orange).

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